OpCode-Stats: A Statistical Analysis of Compiled Program Instruction Sequences

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***Abstract*—Compiled binaries, viewed as flat sequences of machine opcodes, exhibit statistical structure analogous to natural language and biological sequences. We present *OpCode-Stats*[1], an open-source toolkit that applies information-theoretic and computational biology techniques to instruction streams. Across multiple corpora (15–40 binaries, 2.9–3.7 M instructions), three independent validation sets, and a 100-binary compiler matrix, we measure opcode frequency distributions, n-gram entropy rates, compressibility, motifs, and manifold dimensionality. Programs occupy a thin structured manifold (d_{95}=8 of 200 bigram dimensions) with strongly Zipfian opcode concentration ($\hat{\alpha}_{\mathrm{MLE}}\approx1.4$); a bigram language model attains perplexity 8.5, within 0.07 bits of the empirical conditional-entropy lower bound. Mutual information decays to half within $\approx$2 steps; a code-clone pipeline confirms 537 clone pairs in 117 families (96.6% alignment-confirmation rate). Systematic ablation shows the structure persists after aggressive boilerplate stripping (max 3.2% change), replicates across corpora, and is compiler-invariant (compiler $R^2<0.03$).**

***Index Terms*—binary analysis, opcode statistics, information theory, language models, compression, Zipf's law, code clones**

I. Introduction

The last decade has witnessed remarkable progress in language models trained on source code [1], [2]. Yet the statistical properties that make code so learnable remain poorly characterised at the machine-instruction level. Source code is parsed, compiled through many transformations, and reduced to a sequence of machine opcodes that discards all high-level structure. Yet, as we show, this opcode stream retains profound statistical regularity. Bioinformatics offers a natural toolkit for such sequences: k-mer analysis, entropy profiling, motif discovery, and information-theoretic metrics [3]. We extend this toolkit to compiled binaries.

Contributions. (i) An open-source analysis toolkit, *OpCode-Stats*, applying computational-biology techniques to compiled binaries. (ii) Empirical characterisation of opcode frequency concentration: heavy-tailed rank-frequency distribution (MLE $\hat{\alpha}\approx1.4$; KS 0.117); a unigram–sequential decomposition shows 69% of compression gain over random arises from frequency skew, 31% from sequential structure. (iii) Quantification of n-gram entropy and deviation from a unigram-shuffled baseline (gap 1.12 bits at n=5), revealing multi-scale dependencies. (iv) Systematic motif discovery linking recurrent k-mers to specific compiler-generated idioms; positional analysis of function boundaries. (v) A thin-manifold characterisation via compression, mutual information decay, and corpus-level PCA (d_{95}=8 of 200 bigram dimensions). (vi) A controlled compiler matrix (5 projects $\times$ {gcc,clang} $\times$ {-O0,-O2,-O3}) showing source identity and compiler identity are comparably strong determinants of similarity (both $p<10^{-4}$, $r\approx0.5$; project vs. compiler: p=0.38); compiler variation adds new manifold dimensions (d_{95} doubles from 8 to 17). (vii) Empirical validation that n-gram language model perplexity tracks the conditional-entropy lower bound (bigram gap $<$0.07 bits). (viii) A MinHash+LSH+Smith–Waterman clone pipeline that confirms 96.6% of LSH candidates as clones, mapping the manifold to a small catalogue of clone archetypes.

The rest of the paper is organised as follows. Section II reviews related work. Section III describes our pipeline. Section IV presents empirical findings. Section V reports validation and robustness checks. Section VI interprets results in the context of language model performance. Section VII concludes.

II. Background and Related Work

A. Opcode-Level Program Analysis

Binary analysis for security and malware detection has a long history of exploiting opcode frequency features [4], [5]. Bilar [4] observed Zipf-like distributions in Windows PE files; subsequent work used n-gram opcode features for malware classification [5], [6], focusing on discriminative accuracy rather than information structure. The closest prior cross-domain study is Gan et al. [7], who applied n-tuple Zipf analysis to DNA, English, source, and binary. We extend this comparison with mutual-information decay, LZ78 complexity, motif discovery, clone detection, and a controlled compiler experiment. In the malware domain, MAlign [8] encodes raw bytes as nucleotides for genome-style alignment, demonstrating the practical value of the biological-sequence analogy for binary analysis.

B. Binary Embeddings and Similarity

Recent binary code embedding systems implicitly validate the existence of statistical structure in assembly but do not directly characterise it. PalmTree [9] learns BERT-style instruction embeddings; jTrans [10] extends this with jump-aware Transformers for binary similarity detection. Marcelli et al. [11] provide the most comprehensive systematisation, evaluating approaches across 243 K binaries and 26.8 M functions; their benchmark is the natural target for future large-scale validation

[1] https://github.com/OpCode-Stats/OpCode-Stats

of OpCode-Stats's similarity modules. HermesSim [12] argues that graph-based semantic representations outperform NLP-style n-gram approaches for binary *similarity search*; this is orthogonal to our focus on statistical characterisation. Nova [13], a generative language model for assembly, similarly demonstrates learnability but not measurement.

C. Compression and Information Theory

Normalized Compression Distance (NCD) [14] provides a parameter-free measure of similarity grounded in Kolmogorov complexity. Its approximation via `zlib` or `lzma` has been applied to clustering music, genomes, and source files. We apply NCD systematically to binary instruction sequences and compare it against n-gram cosine similarity. Shannon entropy and mutual information have been applied to source code to measure predictability [15], [16]. The *naturalness* hypothesis [15] states that software is more predictable than natural language because it is written by humans following conventions; cross-entropy of token-level language models on code is lower than on English prose. Our work extends this line to the binary level, measuring structural properties that survive compilation.

D. Motif Discovery in Sequences

K-mer motif discovery originated in genomics for finding transcription factor binding sites [17]. In the binary domain, compiler idiom recognition has been studied in the context of decompilation and binary lifting [18], but systematic statistical characterisation of recurrent opcode patterns is absent from the literature.

III. Methodology

A. Corpora and Extraction

We analyse two corpora of GNU/Linux ELF x86-64 binaries from `/usr/bin`: *Smoke* (15 binaries: grep, sed, gzip, bzip2, xz, find, tar, bash, python3, curl, git, vim, nano, gcc, objdump) and *Coreutils-system* (40 binaries spanning 11 functional categories, $\leq$ 10 MB each). The latter adds awk, diff, zip, unzip, cp, mv, rm, chmod, chown, df, wget, ssh, scp, nc, dash, clang, as, ld, perl, ar, strace, nm, readelf, addr2line to the smoke set. Sequences are extracted via `objdump -d` (60 s timeout per binary); only the mnemonic of each instruction is retained, yielding $s=(o_1,\ldots,o_N)$ over a shared vocabulary $\mathcal{V}$. Functions are extracted separately to support per-function motif and positional analyses.

The 15 smoke binaries are not a random draw but a stratified one: they span eight functional categories (text processing, compression, file management, shells, interpreters, networking, editors, developer tools) and three orders of magnitude in size (3,450–777,060 instructions), so the corpus exercises diverse code-generation patterns rather than one program family. Crucially, representativeness is *tested* rather than assumed: every headline metric is replicated on the disjoint 40-binary coreutils-system superset, three independent corpora, and a 100-binary compiler matrix (Section V), where the corpus-level estimates move by $\leq$2%. Per-binary tokens are abundant (2.91 M instructions) but non-independent, so we treat N=15 as the effective sample size for corpus-level inference and report this conservatively throughout.

B. Frequency and Zipf

We fit Zipf's law $f(v) \propto r(v)^{-\alpha}$ via OLS on log-log coordinates and via MLE on the truncated discrete Zipf model ($\ell(\alpha)=-\alpha\sum_r f(r)\log r - N\log Z_V(\alpha)$ with $Z_V(\alpha)=\sum_{r=1}^{V} r^{-\alpha}$). 95% CIs come from n=200 bootstrap resamples; goodness-of-fit is reported via the Kolmogorov–Smirnov statistic against the fitted rank-CDF. We treat the KS statistic as a goodness-of-fit indicator, with a rigorous parametric bootstrap p-value left as future work.

C. Entropy and Shuffled Baseline

n-gram entropy $H_n=-\sum_w p(w)\log_2 p(w)$ and entropy rate $h_r(n)=H_n/n$ are computed for $n=1,\ldots,5$. A unigram-preserving shuffle baseline isolates higher-order dependencies; a negative *entropy gain* $\Delta h_r(n)=h_r(n)^{\text{real}}-h_r(n)^{\text{shuf}}$ for $n>1$ confirms genuine multi-opcode structure.

D. Compression and LZ78 Complexity

We measure compression ratios $\rho=|C(s)|/|s|$ under `zlib` and `lzma`, and LZ78 normalised complexity $\tilde{c}=K/\log_2 N$ [19] (where K is the distinct-phrase count). Both are compared to a uniform-random baseline (maximum entropy) and a unigram-shuffled baseline (preserves marginals). The gap between real and shuffled isolates compression gain attributable to *sequential* structure beyond Zipfian skew.

E. Motifs and Positional Analysis

A k-mer motif is a tuple $(o_i,\ldots,o_{i+k-1})$; we enumerate motifs for $k\in\{4,\ldots,12\}$ and retain those covering $\geq$1% of functions. Each motif is annotated with a semantic label (function prologue, loop/branch, call sequence) based on the opcodes present. To characterise function boundary structure, we collect the opcode at each position $0,1,\ldots,W-1$ from the start and end of every function with $\geq 2W$ instructions (W=20). At each position we compute the empirical distribution over $\mathcal{V}$ and its Shannon entropy.

F. Similarity, Manifold, Mutual Information

Pairwise similarity uses NCD $(C(xy)-\min(C(x),C(y)))/\max(C(x),C(y))$ with `zlib`/`lzma`, and L1-normalised n-gram cosine over top-10 k features (no IDF, since small corpora make IDF penalties discard most features). Sliding-window entropy is computed over $W\in\{16,32,64\}$ along each binary's sequence. Mutual information $I(o_t;o_{t+\ell})$ is estimated for lags $\ell=1,\ldots,50$ over a 50 000-instruction prefix; the *MI half-life* is the smallest ℓ^* for which $I(\ell^*)\leq I(1)/2$. Corpus manifold dimensionality d_{95} is the PCA component count explaining 95% of inter-binary variance on top-200 bigram frequency vectors; a Levina–Bickel MLE [20] cross-check uses k=5 nearest neighbours.

TABLE I
CORPUS STATISTICS AFTER INODE DEDUPLICATION.

Corpus	Bins	Cats	Vocab	Instr. (M)
Smoke	15	8	281	2.91
Coreutils-sys	40	11	301	3.74

G. n-gram Language Models

We train Laplace-smoothed n-gram LMs ($n{=}1,\ldots,5$) jointly on an 80/20 token split. We report cross-entropy $H_{\mathrm{LM}}(n){=}-\frac{1}{T}\sum_t \log_2 P(o_t|o_{t-n+1:t-1})$, perplexity $\mathrm{PPL}(n){=}2^{H_{\mathrm{LM}}(n)}$, and the smoothing gap $\delta(n){=}H_{\mathrm{LM}}(n) - (H_n - H_{n-1})$ vs. the empirical conditional-entropy lower bound.

H. Clone Detection Pipeline

A three-stage pipeline. *Stage 1 (MinHash signatures):* each function's opcode sequence is shingled with $k{\in}\{3,4,5\}$ to form a set of k-gram tokens; a MinHash signature of $m{=}128$ permutations is computed, giving a Jaccard estimator with standard error $\leq 1/\sqrt{m}{\approx}0.09$. *Stage 2 (LSH candidate generation):* LSH is applied at Jaccard thresholds $\tau{\in}\{0.70, 0.90\}$ for each k; candidate pairs from the six (threshold, k) configurations are merged. This reduces the comparison budget from $O(N^2)$ to $O(NC)$ with $C{\ll}N$. *Stage 3 (Smith–Waterman alignment):* each candidate undergoes local sequence alignment on opcode sequences, with normalised score $s{=}S/\min(|f_1|,|f_2|)$. Pairs are assigned to mutually exclusive Jaccard bands: Type 1 ($J{\geq}0.95$, exact/near-exact), Type 2 ($0.95{>}J{\geq}0.70$, high overlap), Type 3 ($0.70{>}J{\geq}0.50$, gapped/structurally similar); families are connected components on the clone similarity graph. For each family we compute a *conserved core* (intersection of member opcode sets), a *divergence score* ($1{-}\bar{J}$), and a *cross-binary* flag.

IV. RESULTS

A. Corpus and Vocabulary

After inode deduplication, the smoke corpus contains 15 binaries (2.91 M instructions, $|\mathcal{V}|{=}281$); coreutils-system contains 40 binaries (3.74 M instructions, $|\mathcal{V}|{=}301$), spanning 11 functional categories (Table I). The top-5 opcodes (`mov`, `lea`, `call`, `jmp`, `push`) account for over 50% of instructions, consistent with prior work on x86-64 binaries [4].

Zipf's law. OLS on log-log coordinates yields $\hat{\alpha}_{\mathrm{OLS}}{=}3.34$ (95% CI $[3.05, 3.67]$, $R^2{=}0.92$). MLE on the truncated discrete Zipf model yields $\hat{\alpha}_{\mathrm{MLE}}{=}1.44$ (95% CI $[1.438, 1.440]$, KS 0.117). The two diverge because OLS overweights the long tail of near-zero-frequency opcodes; MLE, properly weighted by counts, lies closer to natural-language $\alpha{\approx}1$. The KS statistic exceeds the asymptotic critical value at $n{=}281$ (0.081), so a strict truncated power law is rejected; the distribution is best characterised as heavy-tailed with approximately Zipfian concentration (Fig. 1).

A residual analysis of the OLS fit shows its high R^2 is misleading. The log-frequency residuals are strongly non-normal

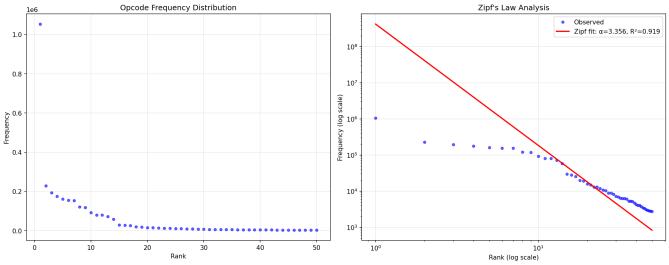

Fig. 1. Opcode rank–frequency distribution (log-log, smoke corpus). The red line is the Zipf fit; the shaded band is the 95% bootstrap CI on α. A handful of mnemonics dominate; the tail decays steeply.

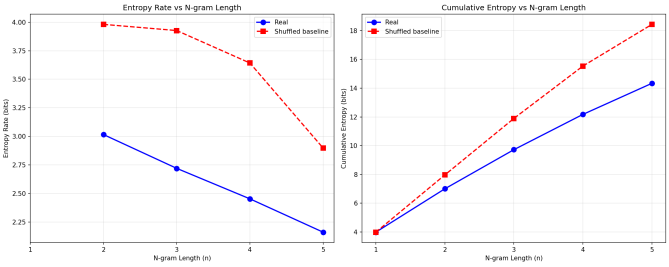

Fig. 2. Entropy rate $h_r(n)$ vs. n for real (blue) and unigram-shuffled (red dashed). The widening gap evidences multi-scale sequential dependencies.

(Shapiro–Wilk $W{=}0.86$, $p{<}10^{-14}$) and highly autocorrelated along rank (Durbin–Watson 0.02, far from the no-structure value of 2): the top-decile and bottom-decile ranks both fall systematically *below* the fitted line (mean residual -0.38 and -0.43 decades) while only mid-ranks are well fit, exactly the head/tail deviation visible in Fig. 1. The residuals therefore reflect structural curvature rather than random scatter, so R^2 does not validate a power law here. This is why we treat the count-weighted $\hat{\alpha}_{\mathrm{MLE}}$—which is insensitive to the unweighted tail—as the reportable exponent throughout.

B. n-gram Entropy

The entropy rate of real sequences decreases by 1.12 bits from $n{=}1$ ($h_r(1){=}3.98$) to $n{=}5$ ($h_r(5){=}2.86$); the shuffled baseline decreases by only 0.30 bits, reflecting near-independence after shuffling. $\Delta h_r(n){<}0$ for all $n{>}1$ confirms genuine multi-instruction dependencies (Fig. 2). Per-binary analysis shows shells and compilers exhibit lower entropy than compression utilities, consistent with predictable control-flow-heavy code vs. data-transform pipelines.

C. Compression and LZ78

Real binaries compress to $\rho_{\texttt{zlib}}{=}0.221$ vs. 0.360 shuffled and 0.666 random; `lzma` gives 0.184/0.303/0.553 (Table II). Decomposing $\Delta{=}0.445$ over random, unigram skew accounts for 0.306 (69%) and sequential order for 0.139 (31%). LZ78 normalised complexity is 630 (median, real) vs. 45,187 (shuffled) and 87,881 (random)—the relative separation, not absolute scale, is the diagnostic since $\tilde{c}$ scales with $N/\log^2 N$. Compression ratio decreases with binary size, but the effect's strength depends on the compressor ($N{=}15$). Under `lzma`

TABLE II
COMPRESSION $\rho = |C(s)|/|s|$ (LOWER IS MORE COMPRESSIBLE).

	Real	Shuffled	Random
`zlib`	0.221	0.360	0.666
`lzma`	0.184	0.303	0.553
LZ78 (norm.)	630 (med.)	45,187	87,881

Fig. 3. Motif score (frequency × function coverage) for the top-5 motifs at each length k. Epilogue and call patterns dominate at all lengths.

the association is strong and significant (Spearman $\rho = -0.87$, $p < 0.001$; Pearson $r = -0.67$, $p = 0.007$); under the smaller-window `zlib` it is weak and not significant (Spearman $\rho = -0.40$; Pearson $r = -0.36$, $p = 0.19$). The significant `lzma` trend indicates larger binaries are more compressible, consistent with greater repetition of library-call patterns; the `zlib` estimate points the same direction but is underpowered at $N = 15$. We prefer Spearman's ρ here because binary size spans three orders of magnitude and the relationship is monotonic rather than linear.

D. Motifs and Compiler Idioms

Motif discovery across $k = 4-12$ identifies 7,623 unique patterns meeting frequency/coverage thresholds; the top-100 per length yield 798 motifs (Fig. 3). High-scoring motifs fall into three semantic categories: (1) *Function epilogues* (highest scores): patterns ending with `pop`+`ret`; the top-ranked 4-mer `pop pop pop ret` appears in 27.9% of all functions. (2) *Call sequences*: patterns centred on `call`, preceded by argument-loading `lea`/`mov`. (3) *Function prologues*: patterns starting with `endbr64` followed by `push` and `mov`, reflecting CET + ABI frame-setup conventions. Remaining patterns comprise loop/branch sequences (`test`+`je`), bulk data-movement, and arithmetic idioms. Coverage at $k = 4$ is highest and declines steadily to $k = 12$, where retained motifs represent compiler-specific idioms with low absolute frequency.

E. Function Boundary Structure

Position 0 of functions is dominated by `endbr64` ($p \approx 0.986$), Intel's CET indirect-branch tracking instruction,

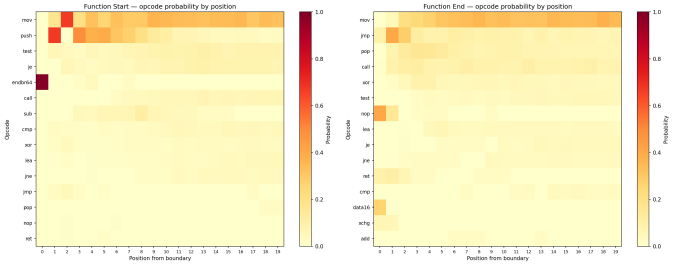

Fig. 4. Opcode probability at function start (left) and end (right). Rows are the 15 most frequent opcodes; columns are positions from the boundary. High-probability cells reveal the CET + ABI prologue convention.

which the Ubuntu 22.04 toolchain inserts at every indirect-call-target function when CET is enabled at the OS level. This is therefore *system-dependent*: on non-CET builds, position-0 would be dominated by `push` (frame setup). At position 1, `push` appears with $p \approx 0.672$; at position 2, `mov` with $p \approx 0.671$—ABI-mandated frame-setup slots that are CET-independent (Fig. 4).

Boundary entropy is ≈ 0.12 bits at position 0—far below the corpus unigram entropy of 3.99 bits—rising to near-corpus levels by positions 5–8. The analysis identifies 3 conserved start positions and 0 conserved end positions ($p > 0.5$), classifying the corpus as *loosely structured*.

F. Similarity, Clustering, and Manifold

The smoke-corpus NCD matrix (`zlib`) ranges 0.956–1.000 (mean 0.991); under `lzma` it ranges 0.852–0.993 (mean 0.967), indicating that `zlib`'s smaller window is in the near-degenerate regime for large binaries (Cebrián et al., 2005); `lzma` should be preferred. Closest pairs are intra-category compression utilities (`bzip2`–`xz`: 0.956; `gzip`–`xz`: 0.963); text-processing tools (`grep`–`sed`: 0.969) are also closer than cross-category pairs.

A one-sided Mann–Whitney U test confirms within-category pairs are stochastically closer than between (`zlib`: $p = 0.013$, permutation $p < 0.001$, $r = 0.600$; `lzma`: $p = 0.005$, permutation $p = 0.003$, $r = 0.698$). Because pairwise distances sharing a binary are not independent, MW p-values are anti-conservative; permutation tests (10,000 label permutations) confirm significance under both compressors.

The absolute gap is small (mean within 0.972 vs. between 0.992 under `zlib`: 13.8% of the between-group baseline), so we interpret NCD as a weak but significant continuous signal, not a basis for hard clustering of heterogeneous corpora.

N-gram cosine similarity over L1-normalised TF vectors decreases with n: 0.953 (bigrams), 0.897 (trigrams), 0.812 (4-grams), with standard deviations 0.037/0.080/0.132. The growing spread is consistent with rare n-grams becoming discriminative as n grows. The UMAP projection of the NCD-`zlib` matrix (Fig. 5) places small compression utilities in a region distinct from large interactive programs (`bash`, `vim`, `git`, `python3`).

Manifold dimensionality. N-gram space coverage drops sharply with n (Table III): even bigrams cover only 6.7% of the theoretical maximum; trigrams 0.15%; 4-grams $\sim 10^{-5}$.

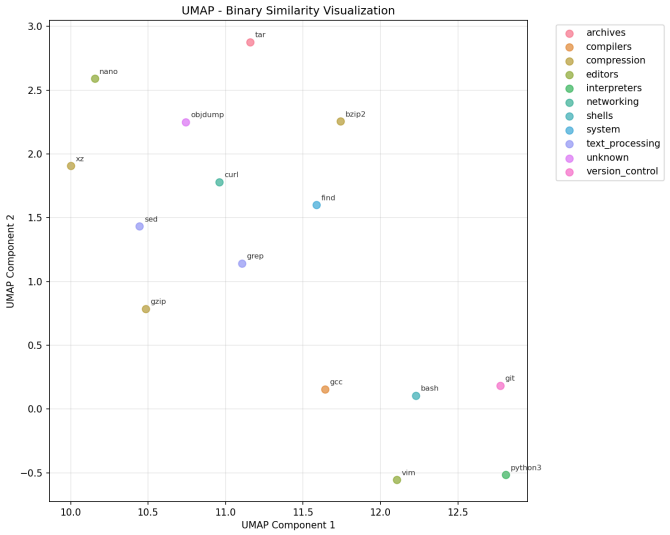

Fig. 5. UMAP projection of the NCD-`zlib` distance matrix (smoke corpus); points coloured by functional category. Compression utilities cluster apart from large interactive programs.

TABLE III
n-GRAM SPACE COVERAGE: UNIQUE OBSERVED VS. THEORETICAL MAXIMUM (SMOKE CORPUS).

n	Unique	Theoretical max	Coverage
1	281	281	100%
2	5,757	85,849	6.71%
3	38,073	25,153,757	0.15%
4	137,534	7.4×10^{9}	1.9×10^{-3}%

Corpus-level PCA on top-200 bigram vectors finds d_{95}=8 of 200 (4%)—a thin structured manifold. A Levina–Bickel MLE [20] cross-check gives $\hat{d}_{\mathrm{MLE}}=8.0\pm5.8$, consistent with the PCA result despite the high variance expected from N=15 points.

G. Mutual Information Decay

$I(1)\approx1.12$ bits, far above the shuffled baseline of ≈0.086 bits, demonstrating that adjacent opcodes share substantial information beyond their marginal distributions. The MI half-life is $\ell^{*}\approx2$ instructions: MI drops below $I(1)/2$ within 2–3 steps, indicating predominantly local structure. The real curve remains above the shuffled baseline at lag 50 ($I(50)$=0.232 vs. 0.086), suggesting weak but non-zero long-range dependencies consistent with repetitive call/ret patterns (Fig. 6).

H. n-gram Language Model Perplexity

Table IV reports the LM evaluation. The unigram LM achieves PPL 19.6 despite a vocabulary of 281—a 15× effective reduction. The bigram LM reaches PPL 8.5 with smoothing gap $\delta(2)$=0.07 bits, within 2% of the conditional-entropy bound; trigram is similarly tight ($\delta(3)$=0.07). For $n\geq4$ the gap grows ($\delta(4)$=0.38, $\delta(5)$=0.97) as Laplace overhead dominates the sparse high-order context space (counts spread

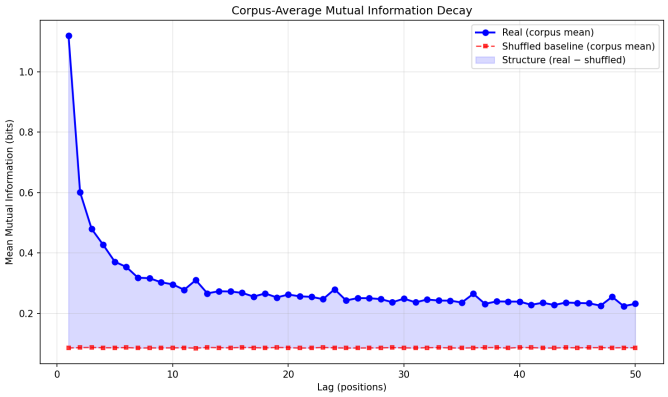

Fig. 6. Corpus-averaged mutual information $I(o_t;o_{t+\ell})$ vs. lag ℓ for real (blue) and shuffled baseline (red dashed). Shaded region is MI attributable to sequential structure.

TABLE IV
n-GRAM LM CROSS-ENTROPY AND PERPLEXITY VS. EMPIRICAL CONDITIONAL-ENTROPY BOUND. LAPLACE SMOOTHING, 80/20 SPLIT.

Model	CE (b)	PPL	Bound (b)	δ (b)
Unigram	4.29	19.6	3.99	+0.30
Bigram	3.08	8.5	3.02	+0.07
Trigram	2.79	6.9	2.72	+0.07
4-gram	2.84	7.1	2.45	+0.38
5-gram	3.13	8.8	2.16	+0.97

over $\sim281^{4}\approx6\times10^{9}$ and $\sim281^{5}\approx2\times10^{12}$ contexts respectively). A back-off or Kneser–Ney model would close much of this gap. The tight n=2, 3 match validates the information-theoretic analysis: corpus-derived entropy rates accurately predict trained-LM perplexity (Fig. 7).

I. Compiler Matrix

We compile five C programs (`sort`, `hash`, `compress`, `search`, `matrix`) under {`gcc`, `clang`} × {−O0, −O2, −O3}, producing 30 binaries. Per-optimisation d_{95}=8 at every level (10 binaries each); the full 30-binary corpus has d_{95}=17, showing compiler/optimisation choice adds new manifold dimensions beyond the source-program span. $-O3$ increases mean instruction count (601 vs. 499 at $-O0$) but also compressibility for compute-intensive code (`matrix` achieves ρ=0.375 via loop unrolling [21]); $-O2$ produces the *least*-compressible output (Table V).

For all $\binom{30}{2}$=435 pairs, one-sided Mann–Whitney tests give within-project $p<10^{-4}$ (r=0.51), within-compiler $p<10^{-4}$ (r=0.55); within-project is *not* significantly tighter than within-compiler (p=0.38, Table VI). Source identity and compiler identity are comparably strong determinants of similarity.

J. Clone Detection

LSH yielded 556 unique candidate pairs on the 6,025 functions of the smoke corpus; Smith–Waterman alignment confirmed 537 (96.6%). These form 117 families, four of which span multiple binaries (Table VII). Type 1 (exact, $J\geq0.95$) dominates (63.7%), consistent with verbatim compiler/linker

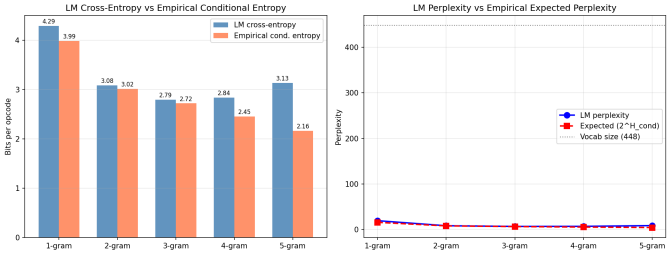

Fig. 7. LM perplexity (blue) vs. expected perplexity $2^{H_n - H_{n-1}}$ (red dashed) and vocabulary size (grey dotted) by n-gram order. Bigram and trigram LMs lie within 0.07 bits of the theoretical lower bound; Laplace overhead dominates at $n \geq 4$.

TABLE V
MEAN INSTRUCTION COUNT AND ZLIB RATIO BY OPTIMISATION LEVEL (AVERAGED OVER 5 PROJECTS × 2 COMPILERS = 10 BINARIES PER ROW).

Opt	Mean instr.	Mean `zlib`
`-O0`	499	0.458
`-O2`	434	0.548
`-O3`	601	0.480

boilerplate (function prologues, `__libc_start_main` wrappers). 5.8% of all functions are in some clone family; the largest family (size 14) is the standard ABI prologue.

Stub-exclusion analysis (below) shows 46% of functions are toolchain-generated, so cross-binary clones include both toolchain-inserted and programmer-authored patterns—the latter a smaller subset. Per-binary clone density varies from 9.6% (`bash`, reflecting its inline library of shell built-in helpers) and 5.0% (`sed`) down to <0.5% for `git` and `vim`, suggesting hand-crafted code with unique control flow is less prone to accidental near-duplication.

V. VALIDATION AND ROBUSTNESS

We conducted a systematic validation programme targeting four threats: extraction errors, boilerplate confounds, corpus dependence, and compiler dependence; full scripts and intermediate data are in `validation/`.

Extraction. An independent reference parser was applied to five binaries (`grep`, `bash`, `find`, `tar`, `xz`); per-mnemonic counts matched exactly (e.g. `bash`: 241,789 instructions). Hard-link deduplication was verified against 7 inode-sharing groups; timeout handling was confirmed via mock injection.

Synthetic and bootstrap checks. Five synthetic corpus families (uniform, Zipf-shuffled, order-2 Markov, templated, mixed) passed all 26 metric checks. Binary-level bootstrap (N=200) gives MLE $\hat{\alpha}$=1.44 [1.36, 1.53], $h_r(5)$=2.86 [2.79, 2.94] bits, mean `zlib` ratio 0.22 [0.18, 0.27]; all CIs exclude reversal. Leave-one-binary-out identifies no influential outlier ($\hat{\alpha}$ varies ≤2%).

Boilerplate ablation. Sixteen ablated variants (strip first/last $k \in \{1, 2, 3, 5\}$ instructions, remove `endbr64`, strip prologues/epilogues, full strip) preserve all metrics within 3.2% (Table VIII). Toolchain functions (startup + stubs) account

TABLE VI
LZMA NCD BY PAIR TYPE ON THE 30-BINARY COMPILER MATRIX. ONE-SIDED MANN–WHITNEY U VS. "BETWEEN".

Pair type	n	NCD	U	p	r
Within-project	75	0.496	5,537	$< 10^{-4}$	0.51
Within-compiler	60	0.539	4,053	$< 10^{-4}$	0.55
Between	300	0.576	—	—	—

TABLE VII
CLONE DETECTION SUMMARY (15-BINARY SMOKE CORPUS, 6,025 FUNCTIONS).

Metric	Value
Functions in any clone family	352 (5.8%)
Confirmed clone pairs	537
Type 1 (exact, $J \geq 0.95$)	342 (63.7%)
Type 2 (high overlap, $0.95 > J \geq 0.70$)	127 (23.6%)
Type 3 (gapped, $0.70 > J \geq 0.50$)	68 (12.7%)
Clone families	117
Cross-binary	4
Largest family (members)	14

for 46% of functions but only 0.6% of instructions; removing them changes mean `zlib` by −0.006 and $h_r(5)$ by +0.003. The findings are intrinsic to application code, not boilerplate. Operand-aware variants (opcode+register class, opcode+immediate bucket) confirm the entropy gap persists.

External validation. Three disjoint corpora pass all pre-registered consistency criteria (Table X): real entropy below shuffled, real compression below shuffled, cross-corpus bigram LM perplexity within 1.2× of within-corpus (range 7.84–9.34). An expanded matrix of 100 binaries ({`gcc`, `clang`}×{`-O0`, `-O1`, `-O2`, `-O3`, `-Os`}×10 programs) gives one-way ANOVA R^2: program identity dominates (R^2=0.40–0.66), compiler identity <3% for all four metrics (Zipf α, H_1, $h_r(2)$, `zlib` ratio), confirming compiler-invariance. Scaling from the 15-binary smoke corpus to the 40-binary coreutils-system superset (Table IX) shows MLE $\hat{\alpha}$, entropy rates, and compression ratios are nearly identical, with manifold dimensionality growing modestly (d_{95}:8→14) as expected when category breadth increases.

Validation summary. Of ten main empirical claims, eight are rated **strong** (CI excludes reversal, ablation < 3.2%, replicates across corpora, compiler-invariant), one **moderate** (cross-binary clones are likely dominated by PLT stubs), one **unsupported** (categorical clustering by binary type: NCD within-vs-between gap is only 13.8% of the between-group baseline).

VI. DISCUSSION

A. The Program Manifold Hypothesis

Compiled programs do not explore opcode-sequence space uniformly. They are confined to a low-dimensional, highly structured manifold shaped by (i) ABI conventions (CET + calling convention impose `endbr64`/push/pop patterns at every function boundary), (ii) compiler optimisation passes (instruction selection, register allocation, peephole optimisation produce characteristic idioms), (iii) algorithm structure (sorting, hashing,

TABLE VIII
BOILERPLATE ABLATION (SMOKE CORPUS). ALL VARIANTS PRESERVE THE ENTROPY GAP AND ZIPF CONCENTRATION.

Variant	$\hat{\alpha}$	$h_r(5)$	**Gap**	`zlib`
Original	1.439	2.863	−0.822	0.221
Strip first 5	1.442	2.867	−0.810	0.224
Strip last 5	1.442	2.866	−0.811	0.224
Strip both 5	1.443	2.866	−0.808	0.224
Remove `endbr64`	1.447	2.862	−0.799	0.225
Full boilerplate strip	1.447	2.865	−0.796	0.225

TABLE IX
METRIC STABILITY ACROSS CORPUS SCALE (SMOKE IS A SUBSET OF COREUTILS-SYSTEM).

Metric	**Smoke (15)**	**Coreutils (40)**
Instructions (M)	2.91	3.74
Vocabulary $\lvert\mathcal{V}\rvert$	281	301
MLE $\hat{\alpha}$	1.439	1.441
$h_r(1)$ (bits)	3.98	3.99
$h_r(5)$ (bits)	2.86	2.88
`zlib` ratio (real)	0.221	0.237
PCA d_{95} (bigram)	8/200	14/200

buffering produce characteristic inner-loop patterns visible as high-coverage motifs), and (iv) clone reuse (linker-inserted stubs and shared libc helpers tile the manifold with exact or near-exact copies). Concentrated vocabulary (MLE $\hat{\alpha}\approx1.4$), rapidly decaying entropy rates, very high compressibility relative to baselines, and $d_{95}\approx8$ collectively show programs are extraordinarily predictable at the instruction level. The validation results (Section V) demonstrate this is not a boilerplate artefact: stripping changes metrics by $\leq3.2\%$, findings replicate across three corpora, and compiler identity explains $<3\%$ of variance.

B. Implications for Binary Analysis

The thin-manifold property has practical consequences beyond statistical curiosity. If programs inhabit a low-dimensional space, similarity measures defined in that space are more informative: two binaries that appear distant in a raw sequence metric may in fact be nearest neighbours on the manifold. Compression-based features provide a useful continuous similarity signal for binary retrieval and provenance without requiring disassembly beyond opcode extraction, but should not be expected to produce clean categorical partitions on heterogeneous corpora. The clone detection results directly confirm this: MinHash + LSH over opcode k-grams recovers 537 clone pairs with a 96.6% confirmation rate, identifying four cross-binary families that correspond to known compiler prologue/libc stubs—without any symbol information or source code.

C. Implications for LMs on Code

The naturalness hypothesis [15] attributes LLM success on source code to its statistical regularity. Our results extend this to the binary level: an LM faces a small effective vocabulary, highly non-uniform transitions, MI decay within

TABLE X
METRIC CONSISTENCY ACROSS THREE INDEPENDENT CORPORA.

Corpus	N	H_1	$h_r(5)$	**Gap**	`zlib`
A (smoke)	15	3.98	2.86	−0.14	0.221
B (net/build)	15	3.97	2.77	−0.21	0.369
C (dev tools)	9	3.99	2.79	−0.17	0.235

≈2 steps, and low intrinsic dimensionality. The bigram LM (Section IV-H) achieves PPL 8.5 within 0.07 bits of the conditional-entropy bound—simple count statistics already capture nearly all bigram-order structure. This $34\times$ effective-vocabulary reduction is determined by the corpus structure alone, not the model choice; it suggests one contributing factor to why learning-based decompilation and lifting methods can recover useful higher-level structure from assembly in practice.

D. Threats to Validity

Construct. Compression conflates frequency skew with sequential redundancy, mitigated but not eliminated by the shuffled baseline. PCA d_{95} reflects variance structure of a frequency matrix, not necessarily geometric manifold dimensionality.

Internal. The boilerplate-driven hypothesis is rejected (Section V); operand-aware variants confirm the entropy gap. Extraction correctness was independently verified with zero mismatches across five binaries.

External. All binaries are x86-64 ELF on GNU/Linux; ARM64, RISC-V, WASM, MSVC, and other source languages are untested. The three independent corpora demonstrate consistency within a single ISA and OS, but cross-platform replication is needed.

Statistical. $N=15$ is a convenience sample; tokens within a binary are non-independent, so corpus-level claims effectively have $N=15$ observations, not millions. Subsampling to $N=4$ preserves directional conclusions but widens CIs substantially. Multiple metrics are reported without formal correction for multiple comparisons.

VII. CONCLUSION

Across a 15-binary smoke corpus (2.91 M instructions), three independent validation corpora, and a 100-binary compiler matrix, opcode frequencies are heavily concentrated (MLE $\hat{\alpha}\approx1.4$), entropy rates decrease by 1.12 bits from $n=1$ to $n=5$ vs. 0.30 shuffled, real sequences compress to 22% vs. 36% shuffled and 67% random, 7,623 unique motifs capture compiler idioms, the manifold occupies $d_{95}=8$ of 200 bigram dimensions, and a bigram LM attains PPL 8.5 within 0.07 bits of the lower bound. Eight of ten empirical claims are strong, one moderate, one unsupported. The thin-manifold hypothesis explains why language models and similarity search are effective on binary code.

Future work. (i) *Multi-ISA replication*: apply the pipeline to ARM64, RISC-V, WASM, and JVM bytecode to test whether Zipf exponents and manifold dimensionality are architecture-independent. (ii) *Operand-level analysis*: extend

opcode sequences to include register classes and immediate ranges, quantifying how operand information changes entropy and manifold structure. (iii) *Compiler / source-language fingerprinting*: distinguish GCC from Clang/MSVC and infer optimisation levels via clustering and motif signatures. (iv) *Transformer-LM evaluation*: extend the n-gram baseline to neural models and quantify additional structure beyond trigrams. (v) *Security applications*: leverage the thin-manifold property for anomaly detection on packed/obfuscated binaries. (vi) *Clone-guided vulnerability propagation*: cross-binary clone families are natural candidates for tracking known-bad function propagation across binaries sharing toolchain or application clones.

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